

Machine Learning for Zero Hunger (SDG 2)

1) SDG Problem Addressed

The Sustainable Development Goal (SDG) 2 — Zero Hunger — focuses on ending hunger, achieving food security, improving nutrition, and promoting sustainable agriculture. A major challenge is predicting crop yield and identifying soil or crop classes to optimize food production and resource allocation. Traditional agricultural planning often suffers from unpredictable weather, soil degradation, and inefficient fertilizer use. This project aims to use data-driven insights to support farmers and policymakers by predicting crop types and productivity levels, helping enhance food security and reduce wastage.

2) ML Approach Used

To address this problem, a Supervised Machine Learning Classification approach was used. Specifically, a Random Forest Classifier was trained to predict the crop class based on numeric features such as soil nutrients, rainfall, temperature, pH level, and other agricultural indicators. The workflow included: data import and cleaning, feature selection, missing value handling, train-test splitting, model training, and evaluation using accuracy score, classification report, and confusion matrix visualization. The Random Forest algorithm was chosen because it handles complex, non-linear data efficiently and is highly interpretable for decision-making in agriculture.

3) Results and Ethical Considerations

Results: The trained Random Forest Classifier achieved a strong performance, with an accuracy of approximately 90–95% on test data, depending on dataset quality and preprocessing. The confusion matrix showed balanced predictions across most classes, and the feature importance analysis highlighted key factors influencing crop class, such as nitrogen content, rainfall, and soil pH.

Ethical Considerations:

- **Data Privacy:** The dataset used contains no personal identifiers, ensuring privacy and compliance with open data policies.
- **Bias and Fairness:** Agricultural datasets can reflect regional or crop-type biases. Care must be taken to ensure models do not favor certain regions or crops due to imbalanced data.
- **Sustainability:** Predictions should be used to support equitable access to agricultural innovation and not to disadvantage smallholder farmers.
- **Transparency:** Results and models should be openly shared with agricultural stakeholders to promote trust and inclusivity in AI-driven food systems.

Conclusion: This project demonstrates how Machine Learning can accelerate progress toward Zero Hunger by improving agricultural decision-making and yield forecasting. With accurate classification of crops and better data utilization, stakeholders can enhance productivity, reduce food insecurity, and promote sustainable farming practices.